E-commerce Conversion Analysis

EXECUTIVE REPORT | Ian Gaffney | BigQuery SQL, Python & Tableau
Google Merchandise Store obfuscated sample data | November 2020 - January 2021

Decision: validate measurement, then test delivery clarity

The analysis does not establish a clear mobile-versus-desktop conversion advantage. It reveals material cart-event gaps and supports a narrower ordered funnel. Validate measurement and investigate checkout behavior before recommending a device-specific redesign.

360,129	1.35%	1,945
Total sessions	Sessions with a purchase	Ordered checkouts without a later same-session purchase*

*December-January mobile/desktop funnel. These are not guaranteed recoverable sales.

What the evidence shows

Device comparison. Mobile exceeded desktop by 0.0776 percentage points. The user-bootstrap 95% interval was -0.0051 to +0.1610 points, including zero. Separate week and first-user-medium adjustments barely changed the estimate. The comparison is observational.

Measurement gaps. The raw-event audit found no mobile/desktop cart events on November 2-15 and November 21-24 despite recorded checkout and purchase activity. The cause is unknown; excluding November does not establish complete tracking.

Restricted funnel. The dashboard selects December-January sessions by session date and device: **49,948 product views > 12,019 ordered carts > 4,246 ordered checkouts > 2,301 purchases**. Ordered checkout completion is **54.19%**. Every stage requires strictly increasing timestamps within one session.

Recommended actions, in priority order

1. Validate event collection. Investigate cart-event gaps and missing earlier-stage events. Distinguish missing records from journeys beginning in prior sessions.
2. Investigate checkout friction. Use usability review and step-level error and exit diagnostics to select interventions.
3. Test credible delivery-date information. A current website review found no arrival estimate for default ground shipping. This is a testable hypothesis, not an explanation of historical behavior.

Proposed experiment and feasibility

Assign eligible users 50/50 before shipping options appear and preserve assignment across visits. Treatment adds a credible arrival range; prices and service speeds stay constant. Measure purchase within seven days per randomized user. Guardrails: revenue per user, shipping mix, errors, cancellations, and delivery accuracy.

At a **hypothetical 50% baseline**, detecting **+2 percentage points** requires about **19,612 users** (80% power; two-sided 5% significance; equal allocation). At 500 new eligible users/day, allow 42 enrollment days plus seven follow-up days. Confirm actual baseline, traffic, and economic value before launch. No experiment has been run.

Scope and traceability

The data are historical and obfuscated; pseudonymous IDs need not represent distinct people across devices. Strict funnels exclude timestamp ties and cross-session journeys. Notebook query 19 filters event dates and yields 2,302 full funnels; dashboard query 20 filters session dates and yields 2,301. Label these populations separately.

Interactive dashboard | SQL, notebook and methodology